

Tut - 06 & 07

Prog 1.

```
#include <stdio.h>
int main() {
    int arr[50];
    int i, n, pos-count, neg-count, zero-count;
    printf("Enter how many numbers  
you wish to input:");
    scanf("%d", &n);
    printf("Enter num\n");
    for (i=0; i < n; i++) {
        scanf("%d", &arr[i]);
    }
    pos-count = 0;
    neg-count = 0;
    zero-count = 0;
    for (i=0; i < n; i++) {
        if (arr[i] > 0)
            pos-count += 1;
        else if (arr[i] < 0)
            neg-count += 1;
        else
            zero-count += 1;
    }
    printf("positive elements = %d\n", pos-count);
    printf("negative elements = %d\n", neg-count);
    printf("There are %d positive elements,  
negative and %d zero", pos-count, neg-count,  
zero-count);
    return 0;
}
```

```

prog2 #include <stdio.h>
int main() {
    int arr[50];
    int i, pos, k;
    printf("Enter numbers");
    scanf("%d", &n);
    printf("Enter numbers (n)");
    for (i=0; i<n; i++) {
        scanf("%d", &arr[i]);
    }

    printf("Enter value of new element to be inserted");
    scanf("%d", &k);
    printf("Enter position of new element : ");
    scanf("%d", &pos);
    for (i=n; i>=pos; i--) {
        arr[i] = arr[i-1];
    }
    arr[pos-1] = k;
    printf("Array after insertion");
    for (i=0; i<=n; i++) {
        printf("%d, arr[i]");
    }

    return 0;
}

```

Q3. #include <stdio.h>
int main() {
 int arr[20];
 int i, n, pos;
 printf("Enter size of array");
 scanf("%d", &n);
 printf("Enter elements in");
 for (i=0; i<n; i++) {
 scanf("%d", &arr[i]);
 }
 printf("Enter position of element to be deleted");
 scanf("%d", &pos);
 for (i=pos; i<n; i++) {
 arr[i] = arr[i+1];
 }
 printf("New array after deletion in");
 for (i=0; i<n-1; i++) {
 printf("%d", arr[i]);
 }
 return 0;
}

Q4. #include <stdio.h>
int main() {
 int arr[50], arr2[50];
 int i, ele, n, choice;
 printf("Enter choice in 1. reverse in 2. store reverse
 in another arr in 3. print reverse arr in");
 scanf("%d", &choice);

switch (choice);

case 1:

printf ("Enter numbers you wish");

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scanf ("%d", &n);

printf ("Enter numbers\n");

for (i = 0; i < n; i++) {

scanf ("%d", &arr[i]);

}

for (i = 0; i < n/2; i++) {

ele = arr[i];

arr[i] = arr[n-i-1];

arr[n-i-1] = ele;

}

for (i = 0; i < n; i++) {

printf ("%d", arr[i]);

}

printf ("\n");

printf ("ARRAY REVERSED");

break;

case 2 :

printf ("Enter numbers");

scanf ("%d", &n);

printf ("Enter numbers");

for (i = 0; i < n; i++) {

scanf ("%d", &arr[i]);

}

for (i = n-1; i >= 0; i--) {

arr2[n-1-i] = arr[i];

printf ("%d", arr2[n-i-1]);

}

printf ("\n");

printf ("Reversed elements of array are stored in
another array");

break;

Cases:

```
printf("Enter how many numbers you want");
scanf("%d", &n);
printf("Enter numbers");
for (i=0; i<n; i++) {
    scanf("%d", &arr[i]);
}
printf("Numbers in reversed order would be: \n");
for (i=n-1; i>=0; i--) {
    printf("%d", arr[i]);
}
break;
}
return 0;
}
```

Q5.

```
#include <stdio.h>
int main() {
    int marks[50], marks2[50], marks3[50];
    int i, j, n, count, highest, lowest, sum, mode;
    float avg, median;
    // store and print marks.
    printf("Enter student marks input");
    scanf("%d", &n);
    printf("Enter marks \n");
    for (i=0; i<n; i++) {
        scanf("%d", &marks[i]);
    }
    printf("The marks are \n");
    for (i=0; i<n; i++)
```

```
{ printf( "%d", marks[i]); }  
printf( "%d\n");
```

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// highest and lowest marks

```
highest = marks[0];  
lowest = marks[0];  
for (i=0; i<n; i++) {  
    if (marks[i] > highest)  
        highest = marks[i];  
    if (marks[i] < lowest)  
        lowest = marks[i];  
}
```

```
printf ("Highest marks %d\n", highest);  
printf ("Lowest marks %d\n", lowest);  
printf ("%d\n");
```

// average

```
sum = 0;  
for (i=0; i<n; i++) {  
    sum += marks[i];  
}  
avg = sum/n;  
printf ("The Average marks is %.f\n", avg);  
printf ("%d\n");
```

// marks in increasing order.

```
for (i=0; i<n; i++) {  
    marks2[i] = marks[i];  
}  
for (i=0; i<n; ++i) {  
    for (j=i+1; j<n; ++j) {  
        if (marks2[i] > marks2[j]) {  
            a = marks2[i];  
            marks2[i] = marks2[j];  
            marks2[j] = a;  
        }  
    }  
}
```

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```

    printf("The numbers in ascending order");
    for (i=0; i<n; i++) {
        printf("%d", marks2[i]);
    }
    printf("\n");

```

```

// median
if (n % 2 == 0)
    median = (marks2[n/2 - 1] + marks2[n/2]) / 2;
else
    median = marks2[n/2 - 1];
printf("The median is %d", median);
printf("\n");

```

```

// mode
int freq = 0;
mode = marks[0];
for (i=0; i<n; i++) {
    count = 1;
    for (j=i+1; j<n; j++) {
        if (marks[i] == marks[j])
            count += 1;
    }
    if (count > freq) {
        freq = count;
        mode = marks[i];
    }
}
printf("mode is %d", mode);
printf("\n");

```

```

return 0;
}

```